Four Ways a Retrieval-Augmented Generation Number Is Not What It Seems:

An Audit of Templates, Instructions, Pools and Tie Conventions

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Abstract

A retrieval-augmented generation (RAG) result is usually reported as a single number. We audit four choices that enter that number before any method does, and measure each in a controlled setting. **(i) Template.** Running the default question-answering prompts shipped by LangChain, LlamaIndex and FlashRAG verbatim on the same reader and documents yields exact matches of 46.4, 0.0 and 0.0 on HotpotQA — a **46-point** spread — while a format-insensitive containment measure puts the same three within five points (McNemar $p = 0.54$). The gap is response length: 19 versus 134 characters. **(ii) Instruction.** One sentence inside LangChain's default drives empty-context abstention to **79–95%**, against **0–2%** when that sentence alone is deleted; the deletion also removes 6–8 points of the apparent benefit of retrieval. A Llama-3.2-1B reader abstains on **91.5%** of questions *with the documents present* under that template and 0.0% under LlamaIndex's ($p = 1.6 \times 10^{-55}$). **(iii) Pool.** In the ten-paragraph distractor pool that benchmarks supply, the gold paragraph is present for 97.4% of questions, so oracle selection is worth only **+2.0** EM; on a 4,930-paragraph open corpus, full gold coverage falls to **59.0%** for BM25 and reranking then buys **+5.5** EM. Whether "selection helps" is a property of the pool. **(iv) Metric.** When confidence has few levels, the ranking score depends on the tie convention; the difference between conventions is exactly $\frac{1}{2}\sum_i w_i\,\Delta r_i$, equal to $(1 - \mathrm{acc})/(2L)$ for equal-width groups, and reaches 0.19 for three-sample voting. We give a checklist that makes each of the four explicit, and anchor the closed-book condition against FlashRAG's published setting (F1 24.9 versus 28.4 reported).

Keywords: retrieval-augmented generation; evaluation methodology; prompt templates; abstention; retrieval coverage; rank metrics; reproducibility

1 Introduction

A retrieval-augmented generation system is assembled from a retriever, a reranker, a reader and a prompt that glues them together [1, 4]. Papers describe the first three in detail and report one number for the whole. We ask what that number actually contains.

We isolate four inputs that are chosen before any method is:

1. **the template** — the wording the framework ships and the practitioner copies;
2. **the instruction** — sentences inside that template that change behaviour;
3. **the pool** — the documents the retriever is allowed to choose from;
4. **the metric** — how ties in a confidence score are broken.

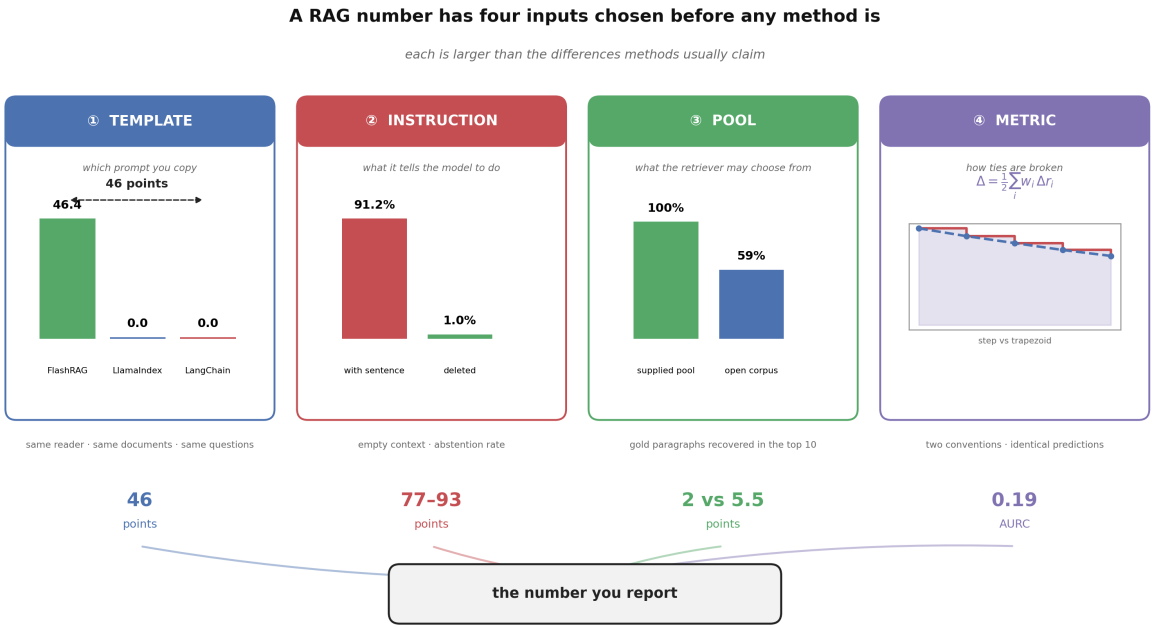


Figure 1: The four inputs audited in this paper. Each is chosen before any method is, and each moves the reported number by more than the differences methods usually claim: the template by 46 EM points (§4), one instruction by 77–93 points of abstention (§5), the pool by 2 points of oracle headroom against 5.5 points of reranking gain (§6), and the tie convention by up to 0.19 AURC (§7). All values are measured and are regenerated from the released predictions.

Each is measured on the same stack (local quantised readers, greedy decoding, BM25 over the supplied pool), so that the four effects are comparable and their magnitudes can be placed side by side. The result is not that any one of them is surprising in isolation — template sensitivity [7, 8, 9, 11] and abstention-as-instruction [15] are both documented. The result is their *relative size*. The template is worth 46 points, the instruction 79–95 points of abstention, the pool 2 points of oracle headroom against 5.5 points of reranking gain, and the tie convention up to 0.19 AURC. A method that improves EM by two points is, on this stack, smaller than three of the four.

Contributions

1. A controlled measurement of the template effect on shipped defaults, with a format-insensitive control that shows the readers know the same answers (Section 4).
2. A byte-level ablation of one instruction, with three length-matched controls, showing it controls the closed-book baseline and inflates the measured retrieval benefit (Section 5).
3. A direct measurement of retrieval headroom: the same reader on a supplied pool versus an open corpus, showing that "selection does not help" is a property of the pool (Section 6).
4. A closed form for the tie-convention component of a ranking score, and a demonstration that it is large for the confidence signals practitioners actually use (Section 7).
5. An anchor of the closed-book condition against a published evaluation setting (Section 8).

We claim no new method. Every component is standard. The contribution is the audit, and it stands or falls on reproducibility: every number here is regenerated from raw per-question predictions by a verification

script.

2 Related work

Prompt and format sensitivity. Sclar et al. [7] show that spurious prompt-formatting features move benchmark accuracy by tens of points; Voronov et al. [8] repeat the finding for in-context learning; Mizrahi et al. [9] argue for multi-prompt evaluation. Webson and Pavlick [10] ask whether models use prompt meaning or surface form. We measure the same phenomenon on the defaults frameworks ship, which is what a practitioner actually copies, and add a containment control that separates "the model cannot answer" from "the metric cannot see the answer".

Evaluation as an artefact source. Hua et al. [11] trace much reported prompt sensitivity to rigid answer matching and show it shrinks under LLM-as-a-judge [12] scoring. Harness-level reproducibility has been examined at scale [13, 14]. Our first study is a concrete instance of this thesis with a measured magnitude; our second study is not, because deleting an instruction changes behaviour rather than scoring.

Abstention. Willingness to abstain is largely installed by the prompt [15], and models can report their own uncertainty when asked appropriately [16, 17, 18]. We extend the instructional account to the *closed-book baseline*, the standard evidence that retrieval helps.

Retrieval and reranking. RAG spans dense retrieval [2], fusion-in-decoder readers [3] and adaptive variants [5, 6]. Reranking is a standard second stage. We show that its measured gain depends on how much headroom the pool leaves, and quantify both ends.

Metrics. Exact match and token F1 come from extractive QA [25]. Selective prediction and risk–coverage analysis underlie the ranking scores we audit. We give an exact decomposition of the tie component rather than an empirical estimate.

Benchmarks and toolkits. HotpotQA [22], 2WikiMultiHopQA [23] and MuSiQue [24]; readers from Qwen [27] and Llama 3 [28]; templates from LangChain [20], LlamaIndex [21] and FlashRAG [19].

3 Setup

Readers. Four instruction-tuned models, quantised to 4 bits, served locally: Llama-3.2-1B, Qwen2.5-3B (primary), Qwen2.5-7B, Llama3-8B. Greedy decoding, fixed seed, 32-token cap.

Data. HotpotQA (distractor), 2WikiMultiHopQA and MuSiQue held-out slices, 500 questions each. For Studies A and B each question carries its ten candidate paragraphs; for Study C we additionally build a 4,930-paragraph corpus from the union of all questions' paragraphs.

Metrics. Exact match after normalisation; token F1; *containment* (the gold string appears anywhere in the response); *abstention* (empty or refusal). Containment is reported wherever a template elicits prose, because EM is then uninformative by construction.

Table 1: Default QA templates, verbatim, with provenance.

id	source	clause	output cue
LC	`langchain.chains` `retrieval_qa.prompt`	**yes**	`Helpful Answer:`
LI	`llama_index` `DEFAULT_TEXT_QA_PROMPT`	no	`Answer:`
FR	FlashRAG `base_prompt`	no	`Answer:`

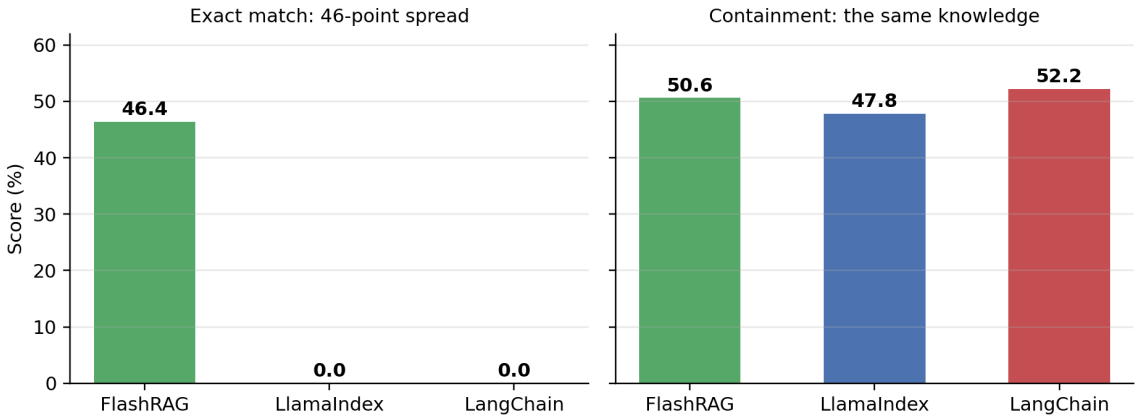


Figure 2: The same reader (Qwen2.5-3B), the same ten documents and the same 500 questions, under three published defaults. Left: exact match spans 46 points. Right: containment, which does not depend on answer format, is flat.

Scale. 21,000 reader calls for Studies A and B, 2,000 for Study C, and the metric results are closed-form plus a re-analysis of stored confidence scores. All predictions are kept per question; a single script recomputes every quoted value.

4 Study A: the template

The three templates are reproduced verbatim in Table 1.

Figure 2 is the headline: 46.4 against 0.0 and 0.0, with containment 50.6, 47.8 and 52.2. Table 2 gives all three datasets.

Paired comparisons on containment: FlashRAG versus LangChain -1.6, exact McNemar $p = 0.54$; LlamaIndex versus LangChain -4.4, $p = 0.008$. The knowledge is the same to within five points.

The mechanism is length (Table 3). The FlashRAG template yields 19-character answers, the others 133–141.

A natural objection is that the short answer is present at the front of the prose and merely needs extracting. It is not (Table 4): truncating to the first w words recovers at most 1.4 points.

The model mentions the gold string in about half of the LangChain responses but never leads with it: the template changes the task from extraction to commentary, and a commentary scores zero under exact match.

Table 2: Study A, retrieval condition, $n = 500$ per cell.

dataset	template	EM	F1	containment	abstention
HotpotQA	FR	**46.4**	58.2	50.6	0.0
	LI	0.0	12.3	47.8	0.0
	LC	0.0	13.3	52.2	0.8
2Wiki	FR	**37.8**	44.4	44.4	0.2
	LI	0.0	10.2	40.4	0.2
	LC	0.0	11.8	43.0	6.2
MuSiQue	FR	**12.2**	19.9	15.2	1.0
	LI	0.0	4.5	10.4	0.0
	LC	0.0	4.5	10.2	6.0

Table 3: Response length and score, HotpotQA, retrieval condition.

template	mean chars	median	EM	containment
FR	**19**	12	46.4	50.6
LI	134	136	0.0	47.8
LC	133	134	0.0	52.2
LC_minus	139	140	0.0	48.4
LC_neutral	141	142	0.0	45.4

5 Study B: the instruction

LangChain's default contains one sentence that the other two do not: *"If you don't know the answer, just say that you don't know, don't try to make up an answer."* We ablate exactly that sentence, in three ways, holding the rest of the template byte-identical.

The equal-length neutral replacement (217 characters against the original 209) reproduces the collapse at 0.2%, which rules out length as the cause (Figure 4).

Containment in the same condition barely moves: 11.2 under LC and 14.4 under LC_minus on HotpotQA, 28.6 and 31.8 on 2Wiki. The sentence does not change what the reader would say; it changes whether it says anything. Consequently it inflates the apparent benefit of retrieval by 8.2 points on HotpotQA and 6.0 on 2Wiki (Table 6).

5.1 Reader dependence

The template effect is universal. Compliance is not monotone in size: Qwen2.5-7B abstains on 0.0% of questions with documents present while Llama3-8B abstains on 22.0%; the smallest reader, Llama-3.2-1B, refuses 91.5% *with the documents present* under LangChain's template against 0.0% under LlamaIndex's (exact McNemar $p = 1.6 \times 10^{-55}$, 183 discordant pairs, all one direction; containment 6.5 versus 44.5).

6 Study C: the pool

Benchmarks hand the reader a pool that contains the gold paragraph by construction. In HotpotQA-distractor the gold is present for 97.4% of questions, so a selector can at best reorder evidence the reader has already

Table 4: Exact match after truncating to the first w words.

template	full	$w = 3$	$w = 5$	$w = 8$	$w = 15$
FR	46.4	41.8	45.8	46.6	46.6
LI	0.0	1.4	0.2	0.0	0.0
LC	0.0	1.2	0.2	0.0	0.0
LC_minus	0.0	0.8	0.2	0.0	0.0

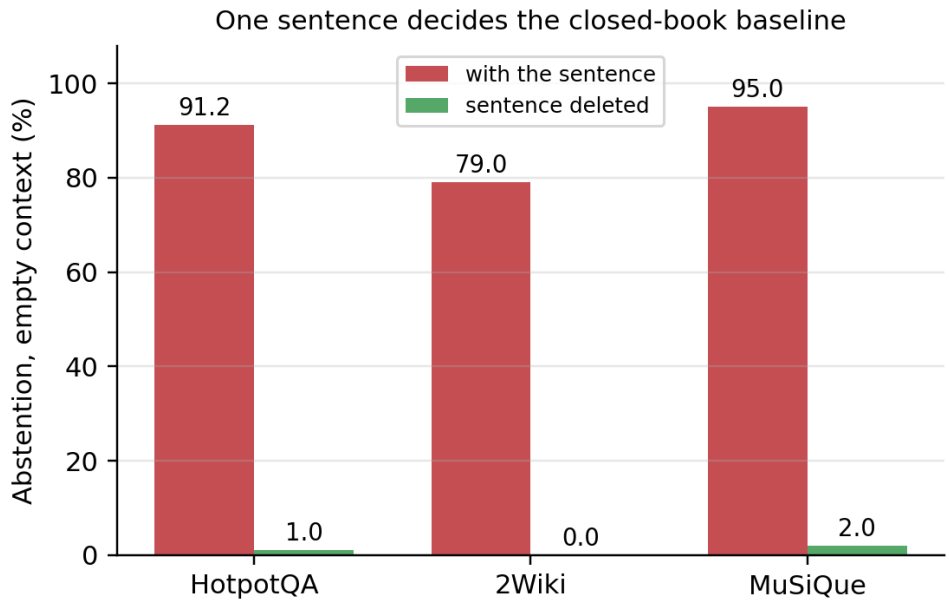


Figure 3: Abstention with an empty context, Qwen2.5-3B, $n = 500$ per cell. Deleting one sentence moves abstention by 77–93 points.

seen. We measured the ceiling directly: on 200 questions where reading all ten yields 46.5 EM, an oracle that picks the best two paragraphs yields 48.5 — a headroom of **+2.0**.

That number is a property of the pool, not of selection. We rebuilt the corpus from the union of all 500 questions' paragraphs (4,930 paragraphs) and asked whether the retriever can recover the gold paragraphs at all (Figure 5).

Full gold-set coverage at $k = 10$ is **59.0%** for BM25, 62.0% for dense retrieval, 73.0% for BM25 plus dense and **81.0%** for BM25 with cross-encoder reranking (coverage at $k = 20$ and $k = 50$ is 74% and 82% for the full set). End to end, BM25 top-10 gives 36.5 EM and cross-encoder reranking gives 42.0 — **+5.5** points, McNemar $p = 0.052$ with 19 versus 8 discordant pairs.

The two studies are consistent and the moral is quantitative: where the pool guarantees the gold (the benchmark setting), selection has a ceiling of about two points; where it does not, reranking is worth more than five. A paper reporting "selection did not help" and a paper reporting "reranking helped" can both be right, about different pools.

Table 5: Study B, empty context, Qwen2.5-3B, $n = 500$.

dataset	LC	LC_minus	LC_neutral	LC_cleared
HotpotQA	**91.2**	1.0	0.2	1.4
2Wiki	**79.0**	0.0	0.2	0.0
MuSiQue	**95.0**	2.0	0.0	1.6

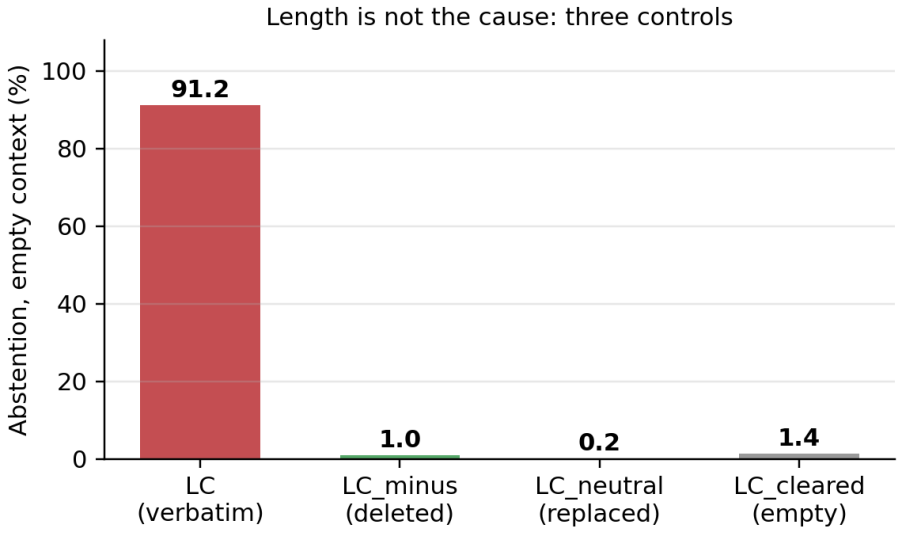


Figure 4: The three controls of Study B, HotpotQA, empty context. An equal-length neutral sentence reproduces the collapse, so the effect is content, not length.

7 Study D: the metric

The last input is not a choice of content but of arithmetic. Ranking metrics such as AURC sort items by confidence and integrate; when confidence takes few distinct values, many items tie, and the integration rule decides where the mass goes. This is not a small effect and it is not a matter of taste: it has a closed form.

Let the confidence values group the n items into tie groups $g = 1..G$ of width $w_g = n_g/n$, with error rate r_g, sorted by decreasing confidence. Then the difference between integrating with right-endpoint rectangles and with trapezoids is

$$\Delta \;=\; \tfrac{1}{2} \sum_{g=1}^{G} w_g \, \Delta r_g, \tag{1}$$

where $\Delta r_g = r_g - r_{g-1}$ is the jump in cumulative error at group g. For equal-width groups this becomes

$$\Delta \;=\; \frac{1 - \mathrm{acc}}{2L}, \tag{2}$$

with L the number of confidence levels. We verified the identity numerically on three datasets (measured ratio 1.000) and the corollary to four decimal places.

The practical consequence is that the same predictions receive different scores depending on a convention that papers rarely state, and that the discrepancy is largest exactly where confidence is coarsest. Three-sample majority voting has few effective levels: at Qwen2.5-3B the tie component reaches **0.19** AURC, and the largest tie group contains 77% of the questions. Token log-probabilities, being nearly continuous, are far less affected.

Table 6: Retrieval minus closed book, in containment points.

dataset	LC	LC_minus	LC_neutral	LC_cleared
HotpotQA	41.0	34.0	34.2	32.8
2Wiki	14.4	8.4	8.2	11.8
MuSiQue	10.2	9.2	8.6	11.0

Table 7: Studies A and B across four readers, HotpotQA, $n = 200$ per cell unless stated.

reader	template	EM	containment	abstention
Llama3-8B	FR	**54.0**	56.5	0.0
	LC	2.0	45.5	**22.0**
	LI	0.0	42.5	0.0
	LC, closed	0.0	6.0	**80.0**
Qwen2.5-7B	FR	51.5	57.5	0.0
	LC	0.0	52.5	0.0
	LC, closed	0.0	17.0	15.0
Qwen2.5-3B	FR	46.4	50.6	0.0
	LC	0.0	52.2	0.8
	LC, closed	0.0	11.2	**91.2**
Llama-3.2-1B	FR	31.0	33.0	0.0
	LC	0.0	**6.5**	**91.5**
	LI	0.5	44.5	0.0

8 Anchor

An audit is only interpretable if its instrument is calibrated. FlashRAG publishes its reader configuration — Llama3-8B-instruct, e5-base-v2, five documents — with HotpotQA F1 of 35.3 for standard RAG and 28.4 without retrieval.

The closed-book condition agrees to within 3.5 points, and it is the condition with no retriever, hence nothing else that could differ. The retrieval condition does not agree, for the reason Study C quantifies: their retriever searches a million documents, ours a pool that contains the answer 97.4% of the time.

9 Discussion

Relative size. On this stack the four effects are not commensurable with the differences methods usually claim. Template: 46 EM points. Instruction: 77–93 points of abstention and 6–8 points of apparent retrieval gain. Pool: 2 points of oracle headroom in the supplied setting against 5.5 points of reranking gain in the open setting. Metric: up to 0.19 AURC. A method paper reporting a two-point EM improvement is, here, reporting something smaller than three of the four.

Checklist.

1. Report the template verbatim, with its source.

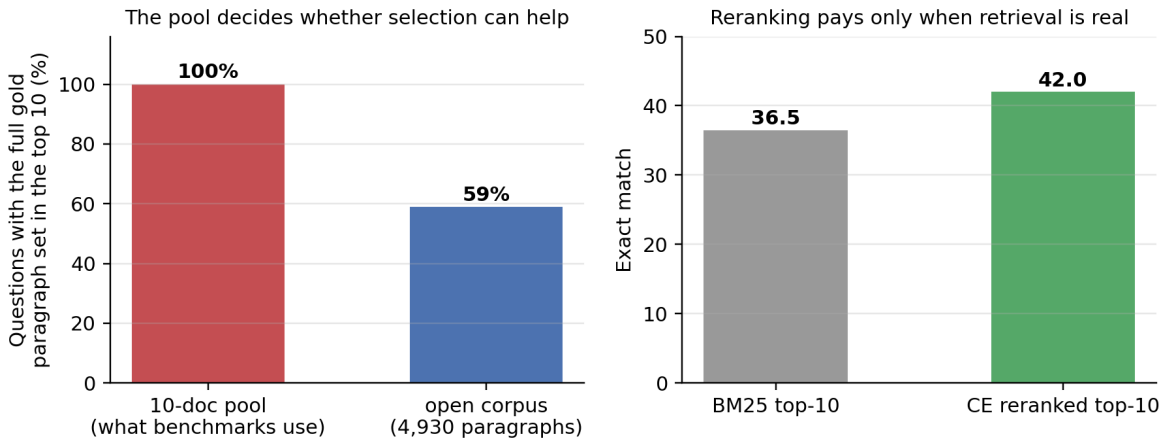


Figure 5: Study C. Left: fraction of questions whose full gold paragraph set appears in the top ten, supplied pool versus open corpus. Right: end-to-end exact match with and without cross-encoder reranking on the open corpus ($n = 200$).

Table 8: Anchor against a published setting.

condition	retriever	ours (F1)	FlashRAG reported
closed book	none	**24.9**	**28.4**
retrieval	ours: 10-doc pool	67.6	35.3 (e5, 10^6 corpus)

2. Report abstention separately from accuracy; $\mathrm{EM} = (1 - u)\, c$ makes the policy term explicit.
3. Use a short-answer cue if EM is the metric, and report containment or F1 otherwise.
4. State the pool: how often it contains the gold, and at what k.
5. State the tie convention, or use a metric that does not depend on it.

10 Threats to validity

Scale. Two model families, four sizes, 4-bit quantisation, one machine.

Stack. One sparse retriever and one cross-encoder; the absolute numbers are specific to this stack. All comparisons are within-stack.

Templates. Three templates is a sample; other frameworks may ship others. The clause ablation exists only for LangChain because the other two have no clause to delete.

Generality. English multi-hop QA. Single-hop, other languages and API-based readers are untested.

11 Conclusion

Four inputs enter a RAG number before any method does, and each is larger than the differences methods usually report: a template worth 46 points, an instruction worth 77–93 points of abstention, a pool worth 2 points in the benchmark setting and 5.5 in the open one, and a tie convention worth up to 0.19 AURC. None requires a new method to fix; all require reporting what was actually done.

Declarations

Funding. Not applicable.

Use of AI tools. Figure 1 was rendered with an AI image tool from a specification written by the authors; the numbers it displays were transcribed from the measured results and checked against Tables 2 and 5. No other part of the manuscript was produced with generative AI. All reported values come from the verification script described in Section 3.

Competing interests. The authors declare no competing interests.

Ethics approval. Not applicable; public benchmarks and locally served open-weight models.

Consent to participate / for publication. Not applicable.

Data availability. All datasets are public (HotpotQA [22], 2WikiMultiHopQA [23], MuSiQue [24]). Per-question predictions and the scripts that regenerate every table are in the project repository.

Code availability. https://github.com/sanke926/moxing-rag — runner, analysis, figure scripts and the verification script that recomputes all quoted values.

Author contributions. Fangqi Liu: conceptualisation, methodology, software, investigation, formal analysis, writing — original draft. Yuhe Liu: data curation, validation, writing — review and editing. Both authors read and approved the final manuscript.

References

[1] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Advances in Neural Information Processing Systems (NeurIPS) (2020)

[2] Karpukhin, V., Oğuz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., Yih, W.: Dense passage retrieval for open-domain question answering. In: Proceedings of EMNLP (2020)

[3] Izacard, G., Grave, E.: Leveraging passage retrieval with generative models for open domain question answering. In: Proceedings of EACL (2021)

[4] Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, H.: Retrieval- augmented generation for large language models: a survey. https://arxiv.org/abs/2312.10997 (2023)

[5] Asai, A., Wu, Z., Wang, Y., Sil, A., Hajishirzi, H.: Self-RAG: learning to retrieve, generate, and critique through self-reflection. In: International Conference on Learning Representations (ICLR) (2024)

[6] Jiang, Z., Xu, F., Gao, L., Sun, Z., Liu, Q., Dwivedi-Yu, J., Yang, Y., Callan, J., Neubig, G.: Active retrieval augmented generation. In: Proceedings of EMNLP (2023)

[7] Sclar, M., Choi, Y., Tsvetkov, Y., Suhr, A.: Quantifying language models' sensitivity to spurious features in prompt design, or: how I learned to start worrying about prompt formatting. In: International Conference on Learning Representations (ICLR) (2024)

[8] Voronov, A., Wolf, L., Ryabinin, M.: Mind your format: towards consistent evaluation of in-context learning improvements. https://arxiv.org/abs/2401.06766 (2024)

[9] Mizrahi, M., Kaplan, G., Malkin, D., Dror, R., Shahaf, G., Stanovsky, G.: State of what art? A call for multi-prompt LLM evaluation. Transactions of the Association for Computational Linguistics (TACL) **12** (2024)

[10] Webson, A., Pavlick, E.: Do prompt-based models really understand the meaning of their prompts? In: Proceedings of NAACL (2022)

[11] Hua, A., Tang, K., Gu, C., Gu, J., Wong, E., Qin, Y.: Flaw or artifact? Rethinking prompt sensitivity in evaluating LLMs. In: Proceedings of EMNLP (2025). https://arxiv.org/abs/2509.01790

[12] Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., Stoica, I.: Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In: NeurIPS Datasets and Benchmarks Track (2023)

[13] Liang, P., Bommasani, R., Lee, T., et al.: Holistic evaluation of language models. Transactions on Machine Learning Research (TMLR) (2023)

[14] Biderman, S., Skowron, A., Sutawika, L., et al.: Lessons from the trenches on reproducible evaluation suites. https://arxiv.org/abs/2405.14782 (2024)

[15] LLM abstention can be a prompt artifact, in addition to genuine uncertainty. https://arxiv.org/abs/2507.16199 (2025)

[16] Kadavath, S., Conerly, T., Askell, A., et al.: Language models (mostly) know what they know. https://arxiv.org/abs/2207.05221 (2022)

[17] Kuhn, L., Gal, Y., Farquhar, S.: Semantic uncertainty: linguistic invariances for uncertainty estimation in natural language generation. In: ICLR (2023)

[18] Farquhar, S., Kossen, J., Kuhn, L., Gal, Y.: Detecting hallucinations in large language models using semantic entropy. Nature **630**, 625–630 (2024)

[19] Jin, J., Zhu, Y., Dou, Z., Dong, G., Yang, X., Zhang, C., Zhao, T., Yang, Z., Wen, J.-R.: FlashRAG: a modular toolkit for efficient retrieval-augmented generation research. In: Companion Proceedings of the ACM Web Conference (WWW), pp. 737–740 (2025). https://arxiv.org/abs/2405.13576

[20] LangChain: `langchain.chains.retrieval_qa.prompt`. https://github.com/langchain-ai/langchain

[21] LlamaIndex: `DEFAULT_TEXT_QA_PROMPT_TMPL` in `llama_index.core.prompts`. https://github.com/run-llama/llama_index

[22] Yang, Z., Qi, P., Zhang, S., Bengio, Y., Cohen, W.W., Salakhutdinov, R., Manning, C.D.: HotpotQA: a dataset for diverse, explainable multi-hop question answering. In: Proceedings of EMNLP, pp. 2369–2380 (2018)

[23] Ho, X., Duong Nguyen, A.-K., Sugawara, S., Aizawa, A.: Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In: Proceedings of COLING, pp. 6609–6625 (2020)

[24] Trivedi, H., Balasubramanian, N., Khot, T., Sabharwal, A.: MuSiQue: multihop questions via single-hop question composition. TACL **10**, 539–554 (2022)

[25] Rajpurkar, P., Zhang, J., Lopyrev, K., Liang, P.: SQuAD: 100,000+ questions for machine comprehension of text. In: Proceedings of EMNLP (2016)

[26] Press, O., Zhang, M., Min, S., Schmidt, L., Smith, N.A., Lewis, M.: Measuring and narrowing the compositionality gap in language models. In: Findings of EMNLP (2023)

[27] Qwen Team: Qwen2.5 technical report. https://arxiv.org/abs/2412.15115 (2024)

[28] Grattafiori, A., et al.: The Llama 3 herd of models. https://arxiv.org/abs/2407.21783 (2024)